

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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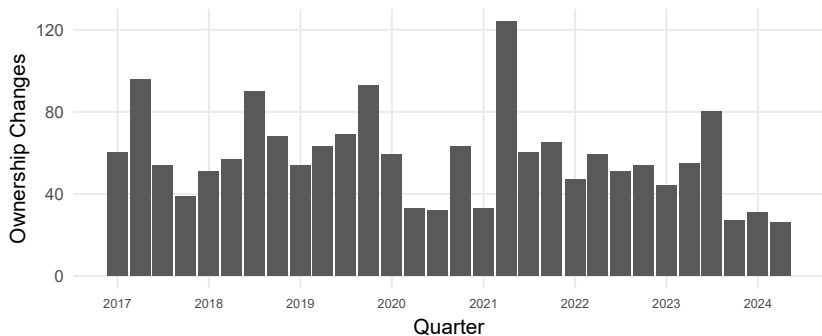
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- This second type of change does **not** alter the set of providers in a market.
- Instead, effects operate through **internal production decisions**.
- This type of change is understudied in the literature, and we examine it in **nursing homes**.

Ownership Changes in Nursing Homes by Quarter



Why Nursing Homes?

- 15,000 U.S. nursing homes serve over 4 million people annually
- 56% of retirement age adults will use a nursing home
- Labor is the main input into care
 - ▶ More staffing improves measured quality ([Lin, 2014](#)) and decreases deficiency citations ([Harrington et al., 2000](#))
 - ▶ Major cost
- Staffing can easily change so it may adjust following an ownership change

Research Question

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 - ▶ Do responses differ across staff type?
 - ▶ Do responses differ if a nursing home is part of a chain or not?
 - ▶ Do responses differ pre vs post-pandemic?

Nursing Staff Types

- **Nursing homes staff care using three main nursing roles:**
 - ▶ **Registered Nurses (RNs):** clinical assessment, care planning, meds, supervision
 - ▶ **Licensed Practical Nurses (LPNs):** routine clinical tasks, meds under RN oversight
 - ▶ **Certified Nursing Assistants (CNAs):** direct daily care
 - ▶ **Total:** RN + LPN + CNA

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 - ▶ **Total:** RN + LPN + CNA
- **Staffing measure: Hours Per Resident Day (HPRD)**
 - ▶ How much total time nurses spend caring for one resident over a 24-hour period
- Ownership change can affect both **levels** and **mix** which sets up cost-quality trade off.

Cost–Quality Trade Off

- Owners can raise quality through two channels:
 - ▶ **Quantity:** increasing total nursing hours (HPRD)
 - ▶ **Skill mix:** shifting hours toward more clinically trained staff
- **Both margins raise costs** because nursing labor is a major operating expense.
- **But short-run revenues are relatively fixed**, so staffing choices map into a **cost–quality tradeoff**.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff. ([Park and Werner, 2011](#); [Bowblis and Brunt, 2025](#))

Why Staffing Could Rise or Fall

- Ownership changes can shift **constraints** and **objectives** $\Rightarrow$ staffing effects are ambiguous.
- **Scenario 1: Cost pressure**
 - ▶ Debt service / tighter targets $\Rightarrow$ cut labor costs
- **Scenario 2: Quality/compliance push**
 - ▶ Invest in staffing to improve oversight / survey performance
- **Scenario 3: Reorganization**
 - ▶ Standardized staffing model shifts RN/LPN/CNA composition; total hours may be similar

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**

- ▶ Annual cost report information
- ▶ Contains information about costs, revenue, financial data and dates of ownership change

- **Nursing Home Compare (NHC) Archive**

- ▶ Contains ownership information as well as facility characteristics

- **Payroll Based Journal (PBJ)**

- ▶ Payroll-reported nurse staffing hours by staff type
- ▶ Used to construct outcome variables

- **Facility-month panel, 48 contiguous states, Jan 2017 – June 2024**

Identifying Ownership Changes

- **Challenge:** ownership is often **layered**, so an administrative “change” may not reflect a true **transfer of economic control**.
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- We identify ownership changes using a **3-step process**.
- **Step 1:** HCRIS includes a variable indicating **whether and when** an ownership change occurred.
 - ▶ Useful for identifying **candidate events**, but the flag may be incomplete or capture legal restructuring rather than an economic change.
- Therefore, we have to use **NHC ownership shares** to verify and date whether control actually changes.

Ownership Level Challenge

- **NHC**

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes

- **Step 2: Validate & date using NHC**

- ▶ Track shares over time within the highest reported ownership level
- ▶ Define a change when **majority control (50%)** transfers between $t - 1$ and t
- ▶ We do not flag changes driven by name changes or within-family reclassification

Identifying Ownership Changes

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● Example from the data:

- ▶ *Before:*
 - ★ ESTES JAMES (84%)
 - ★ BURCHFIELD JOHN (10%)
 - ★ LEE CLAUDE (6%)
- ▶ *After:*
 - ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
 - ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- ▶ We do **not** treat this as a meaningful ownership change

Ownership Change Agreement

- **Step 3: Validate between HCRIS and NHC.**
- We keep facilities only if:
 - ▶ **Never-treated:** 0 changes in **both** NHC and HCRIS, or
 - ▶ **Single change:** exactly 1 change in **each** dataset, within **6 months**.
- **Event month:** first month NHC shows a majority-control transfer.

Cross-tab of NHC & HCRIS

Cross-tab of Ownership Changes

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)
		0	1	2+
		Count of changes in HCRIS		

Dependent Variables: Nurse Staffing (PBJ) and Sample

- We study four staffing measures:
 - ▶ RN, LPN, CNA, and total HPRD
- Outcome is **hours per resident day (HPRD)**: staffing intensity standardized by resident census

$$\text{HPRD}_{s,i,t} = \frac{\text{TotalHours}_{s,i,t}}{\text{ResidentDays}_{i,t}}$$

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- **CMS implausible staffing exclusions:**
 - ▶ RN HPRD = 0 **and** LPN HPRD = 0
 - ▶ Total HPRD < 1.5 or > 12
 - ▶ CNA HPRD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
([Goodman-Bacon, 2021](#))
- **Variation:** within-facility staffing changes relative to facilities not yet treated
- **Event study regression:**

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- **Event study regression:**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- ▶ Y_{it} : staffing outcome in **HPRD**
- ▶ β_{τ} : effect on Y_{it} at event time τ
- ▶ X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- ▶ α_i facility FE; λ_t month FE
- ▶ SEs clustered at the facility and month level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

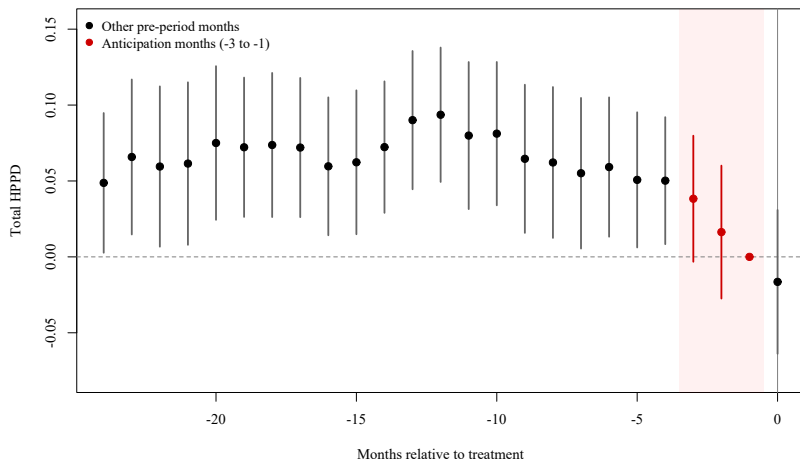
$$Y_{it} = \beta Post_{it} + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- ▶ $Post_{it} = 1$ in all months after a change
- ▶ β : **average post-change effect** on staffing Y_{it}
- ▶ Same fixed effects and controls as the event-study specification
- ▶ SEs clustered at the facility and month level

Identifying Assumptions

- **Parallel trends**
- **Staffing may change in anticipation of a formal ownership change**
 - ▶ A contract is signed, then there is a gap until the closing/transfer date.
 - ▶ During this gap, staffing may adjust
 - ▶ We want to observe the date the contract was signed.
- **Implication:** The months in this window may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period

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- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \beta_\tau = 0 \quad \forall \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

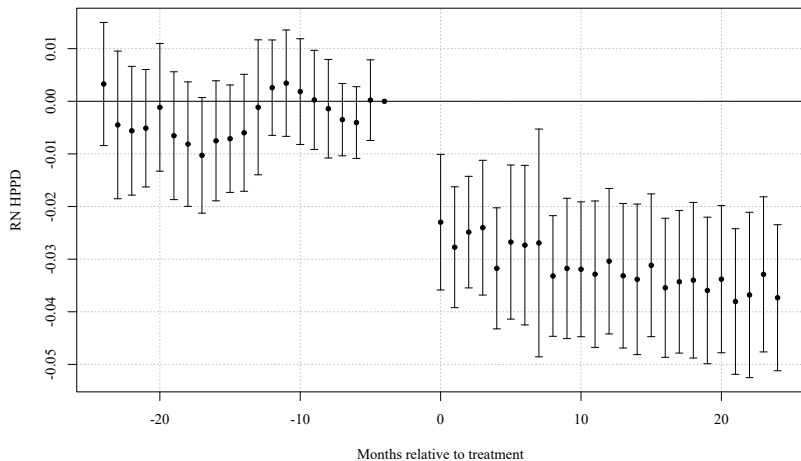
Table: Pre-trend Wald Test p-values (Total Staffing)

	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	0.0006	0.0012
2 Year Window with Donut	0.2046	0.3268
1 Year Window with Donut	0.6387	0.6102

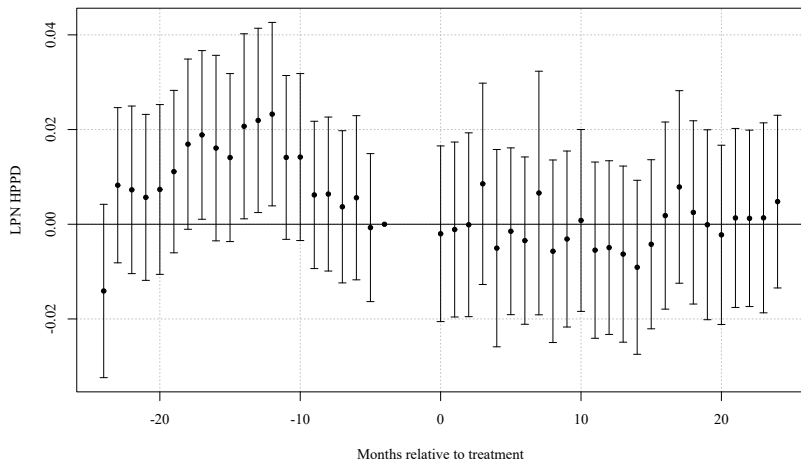
Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.

Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

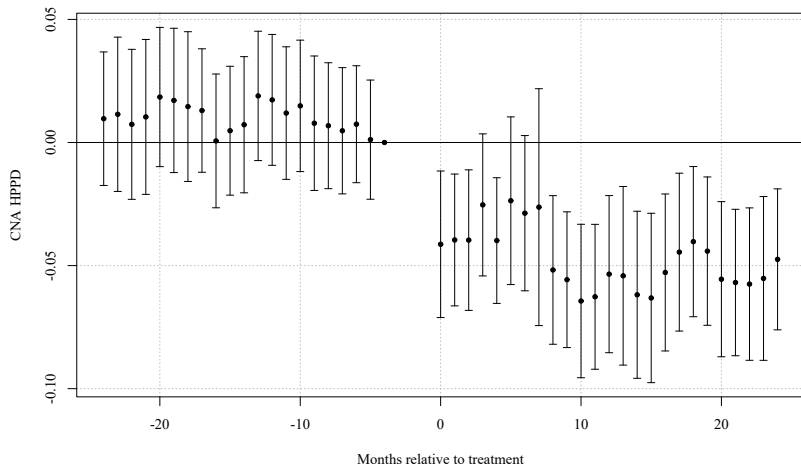
Event Study Results (RN HPRD)



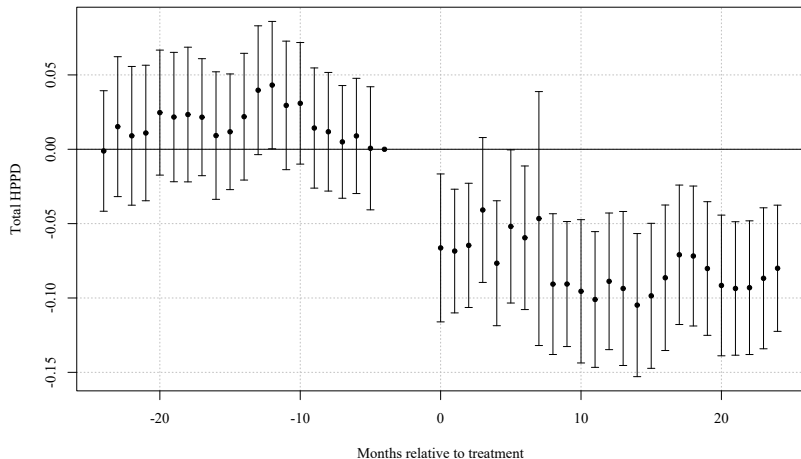
Event Study Results (LPN HPRD)



Event Study Results (CNA HPRD)



Event Study Results (Total Nursing HPRD)



TWFE DiD — Baseline (RN)

	RN	LPN	CNA	Total
HPRD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPRD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628,107$

TWFE DiD — Baseline (LPN)

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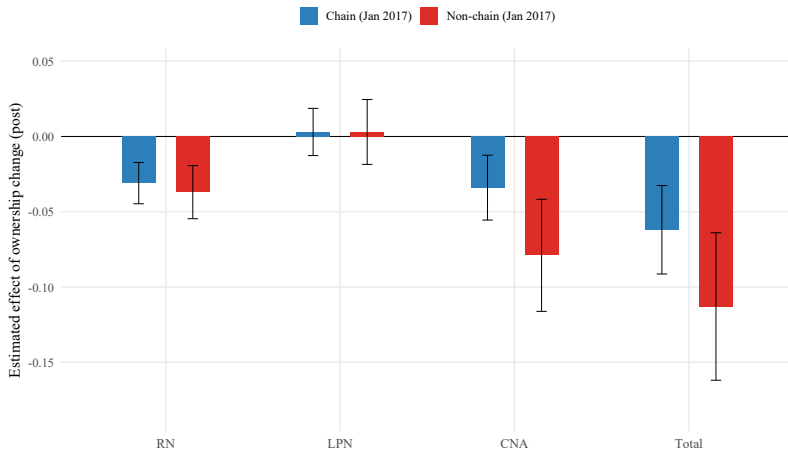
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Heterogeneity: Chain vs Non-Chain Facilities

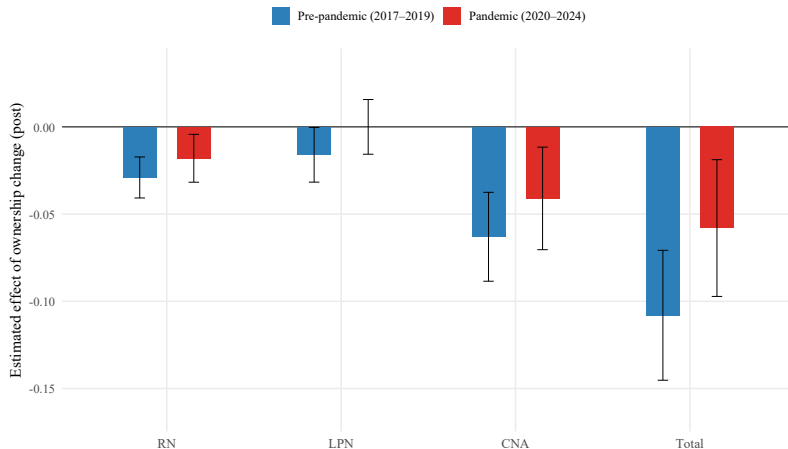
TWFE Post Estimates: Chain vs Non-chain



Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

Heterogeneity: Pre vs Post-Pandemic

TWFE Post Estimates: Pre-pandemic vs Pandemic



Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

Robustness Checks

- **Stacked Difference-in-Difference**
- **Smaller and wider Donut Hole**
- **Gaps in our Data.**

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- **Effects are heterogeneous across settings.**
 - ▶ Larger declines **pre-pandemic** vs. **pandemic period**
 - ▶ Larger declines for **non-chain** vs. **chain** facilities

Discussion

- **Key finding:** RN, CNA, and Total hours fall after ownership change; **LPN hours do not.**
- **Why might LPN hours not decline?**
 - ▶ Compliance
 - ▶ Cost
 - ▶ Substitution

(e.g., Lin 2014; Harrington et al. 2000; Bowblis & Ghattas 2017)

Discussion

- **Question:** Are our findings economically **meaningful**?

Staff type	Sample Average	Effects
RN	25.6 min/day	−1.9 min/day
LPN	47.8 min/day	0.0 min/day
CNA	2 hr 6 min/day	−3.2 min/day
Total	3 hr 19 min/day	−5.2 min/day

- [Lin \(2014\)](#) finds 0.3 HPRD ($\approx$ 18 minutes) increase in RN staffing $\rightarrow$ improves deficiencies 16% .
 - ▶ Our results would translate to a 1.6% increase in deficiencies.

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- **Why are declines smaller after Covid?**

- ▶ Labor market shocks
- ▶ Higher oversight/compliance
- ▶ Tighter constraints

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 - ▶ Identifying mechanisms

Thank you

Questions?

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